

Methods of Proof

Faculty Member, Dept. of CSE, IST

What is a Proof in Discrete Mathematics?

- A **proof** is a series of statements intended to demonstrate some conclusion. Each step in a proof must follow logically from previous steps in accordance with recognized rules of logic and mathematical reasoning.
- In **discrete mathematics**, proofs are central because we deal with statements about integers, sets, logic, graphs, and algorithms, which all require **rigorous reasoning**.
- A proof typically starts with **known facts (axioms, definitions, or previously proven theorems)** and proceeds step by step using **rules of logic and inference** until the desired statement is shown to be true.
- A proof is a valid argument that establishes the truth of a theorem.

Key Terminology Related to Proofs

- **Statement / Proposition:** A declarative sentence that is **either true or false**, but not both.
 - 2 is an even number
- **Axiom / Postulate:** A basic **self-evident truth** that is accepted without proof.
 - Every integer is either even or odd.
- **Theorem :** A **statement that has been proven** to be true based on axioms, definitions, and earlier theorems.
 - The square root of 2 is irrational.

Continue...

- **Lemma** : A **helping theorem** – a small proven result used to prove a bigger theorem.
 - If n is an odd integer, then $n+1$ is also odd.
- **Corollary** : A corollary is a theorem that can be established directly from a theorem that has been proved.
 - **Theorem**: If n is odd, then $n+1$ is even.
 - **Corollary**: If n is even, then $n+1$ is odd.
- **Conjecture** : A statement that is **suspected to be true** based on observations but not yet proven.
 - **Goldbach's Conjecture**: Every even integer greater than 2 can be written as the sum of two prime numbers.

Direct Proof

- A **direct proof** is a method where you start with the **given assumptions (premises)** and use **logical steps** to reach the **conclusion** directly.

Structure:

- If P then Q ($p \rightarrow q$)
 - Assume P is true.
 - Show logically that Q must also be true.

Even + odd = odd

Proposition

- Sum of an even integer and an odd integer is odd.

Proof

- Suppose a is even and b is odd. Then

$$a + b$$

$$= (2m) + b \quad (\text{defn. of even, } a = 2m \text{ for integer } m)$$

$$= (2m) + (2n + 1) \quad (\text{defn. of odd, } b = 2n + 1 \text{ for integer } n)$$

$$= 2(m + n) + 1 \quad (\text{taking 2 as common factor})$$

$$= 2p + 1 \quad (p = m + n \text{ and addition is closed on integers})$$

$$= \text{odd} \quad (\text{defn. of odd})$$

Problems for practice

Prove the following propositions:

- $\text{Even} + \text{even} = \text{even}$
- $\text{Even} + \text{odd} = \text{odd}$
- $\text{Odd} + \text{odd} = \text{even}$
- $\text{Even} \times \text{integer} = \text{even}$
- $\text{Odd} \times \text{odd} = \text{odd}$

n is odd $\Rightarrow n^2$ is odd

Proposition

- The square of an odd integer is odd.

Proof

- **Prove:** If n is odd, then n^2 is odd.

n is odd

$$\Rightarrow n = (2k + 1) \quad (\text{defn. of odd, } k \text{ is an integer})$$

$$\Rightarrow n^2 = (2k + 1)^2 \quad (\text{squaring on both sides})$$

$$\Rightarrow n^2 = 4k^2 + 4k + 1 \quad (\text{expanding the binomial})$$

$$\Rightarrow n^2 = 2(2k^2 + 2k) + 1 \quad (\text{factoring 2 from first two terms})$$

$$\Rightarrow n^2 = 2j + 1 \quad (\text{let } j = 2k^2 + 2k)$$

$(j \text{ is an integer as mult. and add. are closed on integers})$

$$\Rightarrow n^2 \text{ is odd} \quad (\text{defn. of odd})$$

Odd = difference of squares

Proposition

- Every odd integer is equal to the difference between the squares of two integers

Workout

- Write a formal statement.

$\forall$ integer k , $\exists$ integers m, n such that
 $(2k + 1) = m^2 - n^2$.

- Try out a few examples.

$$1 = 1^2 - 0^2$$

$$-1 = 0^2 - (-1)^2$$

$$3 = 2^2 - 1^2$$

$$-3 = (-1)^2 - (-2)^2$$

$$5 = 3^2 - 2^2$$

$$-5 = (-2)^2 - (-3)^2$$

$$7 = 4^2 - 3^2$$

$$-7 = (-3)^2 - (-4)^2$$

- Find a pattern.

$$(k + 1)^2 - k^2 = (k^2 + 2k + 1) - k^2 = 2k + 1 = \text{odd}$$

Odd = difference of squares

Proposition

- Every odd integer is equal to the difference between the squares of two integers.

Proof

- Any odd integer can be written as $(2k + 1)$ for some integer k .
- We rewrite the expression as follows.

$$2k + 1$$

$$= (k^2 + 2k + 1) - k^2 \quad \text{(adding and subtracting } k^2 \text{)}$$

$$= (k + 1)^2 - k^2 \quad \text{(write the first term as sum)}$$

$$= m^2 - n^2 \quad \text{(set } m = k + 1 \text{ and } n = k \text{)}$$

The term m is an integer as addition is closed on integers.

- So, every odd integer can be written as the difference between two squares.

If $a|b$ and $b|c$, then $a|c$

Proposition

- (Transitivity) For integers a, b, c , if $a|b$ and $b|c$, then $a|c$.

Proof

- **Formal statement.**

$\forall$ integers a, b, c , if $a|b$ and $b|c$, then $a|c$.

- c

$$= bn \quad (b|c \text{ and definition of divisibility})$$

$$= (am)n \quad (a|b \text{ and definition of divisibility})$$

$$= a(mn) \quad (\text{multiplication is associative})$$

$$= ak \quad (\text{let } k = mn \text{ and multiplication is closed on integers})$$

$$\implies a|c \quad (\text{definition of divisibility and } k \text{ is an integer})$$

Summation

Proposition

- $1 + 2 + 3 + \cdots + n = n(n + 1)/2$.

Proof

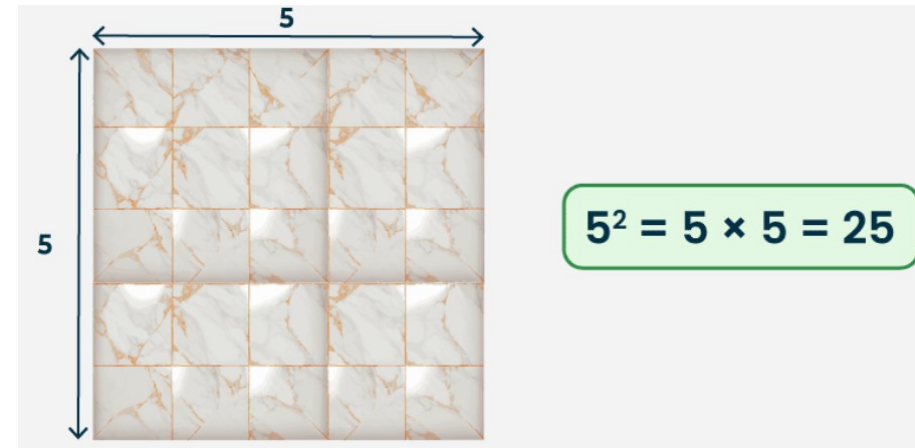
- **Formal statement.** $\forall$ natural number n , prove that $1 + 2 + 3 + \cdots + n = n(n + 1)/2$.
- $S = 1 + 2 + 3 + \cdots + n$
 $\implies S = n + (n - 1) + (n - 2) + \cdots + 1$
(addition on integers is commutative)
 $\implies 2S = \underbrace{(n + 1) + (n + 1) + (n + 1) + \cdots + (n + 1)}_{n \text{ terms}}$
(adding the previous two equations)
 $\implies 2S = n(n + 1)$ (simplifying)
 $\implies S = n(n + 1)/2$ (divide both sides by 2)

Perfect Squares

- A **perfect square** or square number is a number that can be expressed as the product of an integer with itself.
- A number is a **perfect square** if it can be expressed as n^2 , where "n" is an integer.

Example:

- *9 is a perfect square as we can write it as 3×3*
- *16 is a perfect square as we can write it as 4×4*
- *18 is NOT a perfect square as we cannot write as x^2 for an integer x*



Properties of Perfect Squares

Perfect squares have unique characteristics that differentiate them from other numbers. Some key properties include:

- Perfect Squares are the only numbers that have **odd number of distinct divisors**. For example, 9 has divisors as 1, 3, and 9. 4 has divisors as 1, 2 and 4.
- A perfect square always ends in one of these digits: 0, 1, 4, 5, 6, or 9. It will never end in 2, 3, 7, or 8.

If m and n are both perfect squares, then their product nm is also a perfect square.

Since m and n is a perfect square, there must be an integer a such that,

$$m =$$

$$mn =$$

$$mn =$$

$$mn =$$

Mathematical Induction

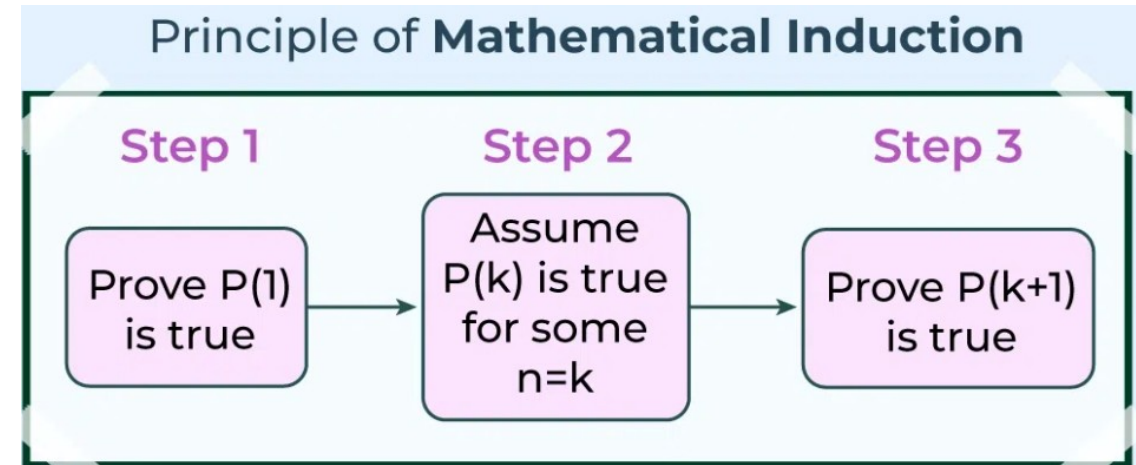
- **Mathematical Induction** is a technique of proving a statement, theorem or formula which is thought to be true, for each and every natural number n .
- Mathematical Induction is a **method of proof** used to show that a statement is true for all natural numbers (usually).
- The principle of mathematical induction is sometimes referred to as **PMI**.

Principle of Mathematical Induction Statement

Base Step: Verify if the statement is true for trivial cases (**$n = 1$**) i.e. check if $P(1)$ is true.

Assumption Step: Assume that the statement is true for $n = k$ for some $k \geq 1$ i.e. $P(k)$ is true

Induction Step: If the truth of $P(k)$ implies the truth of $P(k + 1)$, then the statement $P(n)$ is true for all **$n \geq 1$** .



1. Show that the sum of the first n positive integers: $1 + 2 + \dots + n = n(n+1)/2$

Base Case ($n = 1$)

$$1 = \frac{1(1+1)}{2} = \frac{2}{2} = 1$$

So the statement is true for $n = 1$.

Inductive Hypothesis

Assume the statement is true for some arbitrary positive integer k :

$$1 + 2 + 3 + \dots + k = \frac{k(k+1)}{2}$$

Continue...

Inductive Step (prove for $k + 1$)

Consider the sum up to $k + 1$:

$$1 + 2 + 3 + \cdots + k + (k + 1)$$

Using the inductive hypothesis (IH) for the first k terms:

$$\begin{aligned} &= \frac{k(k + 1)}{2} + (k + 1) \\ &= \frac{k(k + 1) + 2(k + 1)}{2} \\ &= \frac{(k + 1)(k + 2)}{2} \end{aligned}$$

Which is exactly the formula with $n = k + 1$.

2. Show that the sum of the first n positive odd integers is : $1 + 3 + 5 + \dots + (2n - 1) = n^2$

Step 1: Base Case ($n = 1$)

When $n = 1$:

- Left side: The first odd integer = 1
- Right side: $1^2 = 1$

Since $1 = 1$, the statement is true for $n = 1$. ✓

Step 2: Inductive Hypothesis

Assume the statement is true for $n = k$, where k is some positive integer:

$$1 + 3 + 5 + \dots + (2k-1) = k^2$$

Step 3: Inductive Step

We need to prove the statement is true for $n = k+1$:

$$1 + 3 + 5 + \dots + (2k-1) + (2(k+1)-1) = (k+1)^2$$

Starting with the left side:

- The $(k+1)$ th odd integer is $2(k+1)-1 = 2k+1$

So we have:

$$1 + 3 + 5 + \dots + (2k-1) + (2k+1)$$

By the inductive hypothesis, we know that $1 + 3 + 5 + \dots + (2k-1) = k^2$

Therefore:

$$= k^2 + (2k+1)$$

$$= k^2 + 2k + 1$$

$$= (k+1)^2$$

This matches the right side! ✓

Conclusion

By mathematical induction, the statement $1 + 3 + 5 + \dots + (2n-1) = n^2$ is true for all positive

To construct the proof, let $P(n)$ denote the proposition: $n^3 - n$ is divisible by 3.

1. **Base Case:** For $n = 1$,
 $1^3 - 1 = 0$. Since $3 \mid 0$, $P(1)$ is true.
2. **Inductive Hypothesis:** Assume $P(k)$ is true for some integer $k \geq 1$.
Assume $k^3 - k = 3m$ for some integer m .

3. **Inductive Step:** Show $P(k+1)$ is true.

Consider $(k+1)^3 - (k+1)$:

$$= (k^3 + 3k^2 + 3k + 1) - (k + 1)$$

$$= k^3 + 3k^2 + 2k$$

$$= (k^3 - k) + 3k^2 + 3k$$

$$= 3m + 3k^2 + 3k$$

$$= 3(m + k^2 + k)$$

Since $(m + k^2 + k)$ is an integer, $(k+1)^3 - (k+1)$ is divisible by 3. Thus, $P(k+1)$ is true.

3. Use mathematical induction to show that: $P(n) : 1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1$
for all nonnegative integers n .

1. Base Case ($n = 0$)

Since we must prove this for all **non-negative integers**, we start with $n = 0$.

The sum on the Left-Hand Side (LHS) includes terms up to 2^n , so for $n = 0$, the sum is just the first term, 2^0 .

LHS: $1 + 2 + 2^2 + \dots + 2^0 = 2^0 = 1$

RHS: $2^{n+1} - 1 = 2^{0+1} - 1 = 2^1 - 1 = 2 - 1 = 1$

Since LHS = RHS, the statement $P(0)$ is **true**.

2. Inductive Hypothesis

Assume that the statement $P(k)$ is true for some arbitrary **non-negative integer k** .

This means we assume:

$$1 + 2 + 2^2 + \dots + 2^k = 2^{k+1} - 1$$

3. Inductive Step ($n = k + 1$)

We assume the Inductive Hypothesis (I.H.): $1 + 2 + \cdots + 2^k = 2^{k+1} - 1$.

We must show that $1 + 2 + \cdots + 2^k + 2^{k+1} = 2^{k+2} - 1$.

$$\begin{aligned}\text{LHS} &= \underbrace{(1 + 2 + \cdots + 2^k)}_{\text{Using I.H.}} + 2^{k+1} \\ &= (2^{k+1} - 1) + 2^{k+1} \\ &= 2 \cdot 2^{k+1} - 1 \\ &= 2^{1+(k+1)} - 1 \\ &= 2^{k+2} - 1\end{aligned}$$

Since $\text{LHS} = 2^{k+2} - 1$, which is the RHS of $P(k + 1)$, the statement $P(k + 1)$ is **true**.

4. Use the mathematical induction to show the formula $P(n) : \sum_{i=0}^n ar^i = a + ar + ar^2 + \dots + ar^n = \frac{a(r^{n+1} - 1)}{r - 1}$

Step 1: Base Case $n = 0$

For $n = 0$:

$$\sum_{i=0}^0 ar^i = ar^0 = a$$

Right-hand side (RHS):

$$\frac{a(r^{0+1} - 1)}{r - 1} = \frac{a(r - 1)}{r - 1} = a$$

Step 2: Induction Hypothesis

Assume the formula is true for $n = k$, i.e.,

$$\sum_{i=0}^k ar^i = \frac{a(r^{k+1} - 1)}{r - 1}$$

This is our **induction hypothesis**.

3. Inductive Step ($n = k + 1$)

We need to show: $\sum_{i=0}^{k+1} ar^i = \frac{a(r^{k+2}-1)}{r-1}$.

Start with the Left-Hand Side (LHS) and apply the **Inductive Hypothesis (I.H.)**, $\sum_{i=0}^k ar^i = \frac{a(r^{k+1}-1)}{r-1}$:

$$\begin{aligned}\text{LHS} &= \sum_{i=0}^k ar^i + ar^{k+1} \\ &= \frac{a(r^{k+1} - 1)}{r - 1} + ar^{k+1}\end{aligned}$$

Combine the terms by finding a common denominator, $(r - 1)$:

$$\begin{aligned}\text{LHS} &= \frac{a(r^{k+1} - 1) + ar^{k+1}(r - 1)}{r - 1} \\ &= \frac{ar^{k+1} - a + ar^{k+2} - ar^{k+1}}{r - 1} \\ &= \frac{ar^{k+2} - a}{r - 1} \\ &= \frac{a(r^{k+2} - 1)}{r - 1}\end{aligned}$$

This equals the Right-Hand Side (RHS) of $P(k + 1)$, thus $P(k + 1)$ is **true**.

Proof by Contradiction(Indirect Proof)

An indirect proof in discrete mathematics, also known as **proof by contradiction**, is a method of proving a statement by assuming the opposite of what you want to prove is true and showing that this assumption leads to a logical contradiction.

How it Works

- 1.State the proposition:** Clearly define the statement (P) you want to prove.
- 2.Assume the negation:** Assume that the negation of the statement ($\neg P$) is true. This is the starting point of the proof.
- 3.Derive a contradiction:** Use logical reasoning, definitions, and previously established theorems to derive a new statement. This new statement should contradict either a known fact, a definition, or the initial assumption itself.
- 4.Conclude the original statement is true:** Since the assumption ($\neg P$) led to a contradiction, the assumption must be false. Therefore, the original statement (P) must be true.

1. Give an indirect proof of the theorem: "If $3n+2$ is odd, then n is odd."

Step 1: Restate the theorem in logic form

Let P : " $3n + 2$ is odd"

Let Q : " n is odd"

We want to prove: $P \rightarrow Q$.

Step 2: Indirect proof setup

An **indirect proof** means we prove the contrapositive, or we assume the negation of the conclusion and show it leads to the negation of the hypothesis.

The contrapositive of $P \rightarrow Q$ is:

$$\neg Q \rightarrow \neg P$$

i.e., "If n is not odd, then $3n + 2$ is not odd."

Since "not odd" means "even", we can restate:

Contrapositive statement:

If n is even, then $3n + 2$ is even.

Step 3: Prove the contrapositive

Assume n is even.

Then $n = 2k$ for some integer k .

Compute $3n + 2$:

$$3n + 2 = 3(2k) + 2 = 6k + 2 = 2(3k + 1).$$

Let $m = 3k + 1$, which is an integer.

Thus $3n + 2 = 2m$, so $3n + 2$ is even.

Step 4: Conclude

We have shown: If n is even, then $3n + 2$ is even.

This is the contrapositive of the original statement.

Therefore, the original statement is true:

If $3n + 2$ is odd, then n is odd.

2. Give a direct proof of the theorem: "If n is odd, then n^2 is odd."

Step 1: Restate the theorem in logic form

Let P : " n is odd"

Let Q : " n^2 is odd"

We want to prove: $P \rightarrow Q$.

Step 2: Indirect proof setup

An indirect proof means we prove the contrapositive:

$$\neg Q \rightarrow \neg P$$

i.e., "If n^2 is not odd, then n is not odd."

Since "not odd" means "even", we restate:

Contrapositive statement:

If n^2 is even, then n is even.

Step 3: Prove the contrapositive

Assume n^2 is even.

Then $n^2 = 2k$ for some integer k .

We want to show n is even.

Suppose, for contradiction, that n is odd.

Then $n = 2m + 1$ for some integer m .

Then $n^2 = (2m + 1)^2 = 4m^2 + 4m + 1 = 2(2m^2 + 2m) + 1$, which is odd.

But this contradicts the assumption that n^2 is even.

Thus, our supposition that n is odd must be false.

Therefore, n is even.

Step 4: Conclude

We have shown: If n^2 is even, then n is even.

This is the contrapositive of the original statement.

Therefore, the original statement is true:

If n is odd, then n^2 is odd.

If $x^2 - 6x + 5$ is even (where x is an integer), then x is odd.

Proof by contrapositive:

- Contrapositive: If x is even, then $x^2 - 6x + 5$ is odd.
- Assume $x = 2k$.
- Then $x^2 - 6x + 5 = (2k)^2 - 6(2k) + 5 = 4k^2 - 12k + 5$.
- Rewrite: $4k^2 - 12k + 4 + 1 = 2(2k^2 - 6k + 2) + 1$, which is odd.
- So the expression is odd when x is even.
- Contrapositive is true, hence original is true.